SELF-SOVEREIGN SOFTWARE FACTORY

PROJECT IDENTITY: S.E.V.R. // OPS-LEVEL DEVOPS STACK

This document outlines the architecture for a high-density, sovereign development environment tailored for systems engineering. The stack prioritizes performance, explicit logic, and local hardware utilization, bypassing the abstractions of mainstream cloud-native providers.

01. Core Forge Layer

COMPONENT:	Forgejo (LTS)
RATIONALE:	Go-based binary, Git-standard, zero-tracking, FOSS.
ROLE:	Code persistence, Cargo Registry, OCI Image Storage.

02. CI/CD Engine: Concourse CI

Moving beyond the linear "Action" model, Concourse provides a visual, resource-based pipeline architecture. Every input and output is explicitly defined as a resource.

Architecture Philosophy

- **Idempotency:** Every build runs in a clean, ephemeral container.
- **Visual Gravity:** A real-time dependency graph for tracking complex Rust compilation paths.
- **Native Integration:** Seamlessly hooks into S.E.V.R. infrastructure via specialized resource types.

```
# Sample Resource Definition for Rust/Cargo
resources:
- name: source-code
  type: git
  source:
    uri: https://forgejo.internal/styrene/omegon.git
    branch: main
```

```yaml
jobs:
- name: build-release
  plan:
  - get: source-code
    trigger: true
  - task: cargo-build
    config:
      platform: linux
      image_resource:
        type: registry-image
        source: { repository: rust, tag: latest }
      run:
        path: cargo
        args: ["build", "--release"]
```

03. Project Tracking: Vikunja

A high-density alternative to Plane, Vikunja aligns with the "Industrial" aesthetic while maintaining a lightweight footprint.

- **Task Relations:** Map complex dependencies between Rust crates and hardware modules.
- **View Flexibility:** Toggle between Kanban for active sprints and Gantt for long-term antenna profile research.
- **Self-Hosted:** Single binary + DB approach mirrors the Forgejo philosophy.

04. Hardware Integration Matrix

To leverage local silicon (M1/M5 Max), the factory utilizes distributed agents.

COMPUTE A:	M5 Max (128GB) - Primary Build Worker
COMPUTE B:	M1 Max (64GB) - Secondary / AI Inference Testing
COMPUTE C:	RTX 3060Ti - GGUF/Candle Quantization CI

> *"The objective is not to replicate the cloud, but to build a factory floor where the engineer owns the machines."*